

#### 4.2.1 Fertility and assisted reproduction

| Learning outcomes                                                                                   | Additional guidance                                                                                                                        |
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| <i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>       |                                                                                                                                            |
| <b>(a)</b> the parts of the male and female urinogenital systems                                    | To include the interpretation of diagrams, photomicrographs and electron micrographs.<br><i>M0.1, M0.2, M1.1, M1.2, M1.8</i>               |
| <b>(b)</b> <b>(i)</b> the histology of the ovaries and testes                                       | To include the interpretation of diagrams, photomicrographs and electron micrographs.<br><i>M0.1, M0.2, M1.1, M1.2, M1.8</i>               |
| <b>(ii)</b> observations of the histology of the ovaries and testes made using the light microscope | <b>PAG1</b><br>HSW4, HSW5, HSW6                                                                                                            |
| <b>(c)</b> the process of gametogenesis                                                             | To include oogenesis and spermatogenesis and relate these to mitosis and meiosis.                                                          |
| <b>(d)</b> the structure of the secondary oocyte and sperm related to their functions               | To include the interpretation of diagrams, photomicrographs and electron micrographs.<br><i>M0.1, M0.2, M1.1, M1.2, M1.8</i>               |
| <b>(e)</b> the role of hormones in gametogenesis                                                    | To include spermatogenesis: FSH, GnRH, LH, testosterone and inhibin<br><b>AND</b><br>oogenesis: FSH, LH, progesterone, oestrogen and GnRH. |
| <b>(f)</b> the role of hormones in the regulation of the menstrual cycle                            | To include LH, FSH, progesterone and oestrogen.                                                                                            |

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| <b>(g)</b> | the process of fertilisation                                                                | To include the importance in restoring chromosome number and introducing genetic variation.                                                                                                                                                                                   |
| <b>(h)</b> | the use of monoclonal antibodies in pregnancy testing                                       | No details of the production of monoclonal antibodies are required.                                                                                                                                                                                                           |
| <b>(i)</b> | a consideration and evaluation of the biological and ethical issues surrounding infertility | To include the production of abnormal sperm, low sperm count and blocked vas deferens, ovulatory disorders, oviduct blockage, endometriosis and anti-sperm antibodies.<br><i>M0.1, M0.2, M1.1, M1.2, M1.8</i>                                                                 |
| <b>(j)</b> | assisted reproduction                                                                       | Treatments and techniques to include intrauterine insemination, IVF, intra-cytoplasmic sperm injection (ICSI), donor sperm insemination, surgical sperm retrieval, ovulation induction, frozen embryo replacements and gamete intra-fallopian transfer.<br>HSW9, HSW10, HSW12 |